Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R, S) / S4 (PT) (R, S) Examination June 2023 (2019 Scheme)

Course Code: EET322

Course Name: RENEWABLE ENERGY SYSTEMS

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks. Marks

1		List two causes and consequences of global warming.	(3)
2		Explain the harmful effects of three pollutants affecting the environment.	(3)
3		Discuss the effect of temperature and insolation on PV cell.	(3)
4		Explain the terms tilt angle and surface azimuth angle of a solar cell.	(3)
5		Differentiate between HAWT and VAWT.	(3)
6		Explain with illustration the power output versus wind speed characteristics of a wind turbine.	(3)
7		List the advantages and limitations of tidal power plants.	(3)
8		Where do you find the occurrence of biofouling? What are the methods to control the same?	(3)
9		Discuss the methods of urban waste to energy conversion.	(3)
10		Explain the working of fuel cell.	(3)

PART B

Answer one full question from each module, each carries 14 marks.

Module I

11	a)	List the various non-conventional energy resources.	(4)
	b)	Explain the advantages and limitation of each non-conventional energy resource.	(10)

OR

12	a)	Discuss the importance of Green power.	(4)
	b)	Explain the various classification of energy resources.	(10)

Module II

13	a)	Describe the various types of solar concentrators.	(9)
	b)	Explain the working of pyranometer.	(5)

OR

14	a)	What is the effect of partial shading on PV cells.	(3)
	b)	With necessary equations and diagrams, explain the characteristics and equivalent circuit of a solar cell.	(11)

Module III

15	a)	Discuss the effect of wind speed and grid conditions in system integration of wind turbines.	(4)
	b)	Discuss the various classification of turbines for small hydro plants.	(10)

OR

16	a)	Explain the wind energy conversion system with variable speed drive.	(4)
	b)	Derive the expression for maximum power extracted by a wind turbine.	(10)

Module IV

17	a)	Describe the principle and various components of tidal power plants.	(4)
	b)	Explain with the help of block diagram, the working of open cycle OTEC plants.	(10)

OR

18	a)	Advantages and limitations of OTEC.	(4)
	b)	Explain with the help of block diagram, the working of closed cycle OTEC plants.	(10)

Module V

19	a)	Explain the factors affecting biogas generation.	(4)
	b)	Discuss the different types of biogas plants.	(10)

OR

20	a)	Explain the necessity of energy storage.	(4)
	b)	Describe the different methods of energy storage.	(10)

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